

SUMMARY TABLE: CORE MATHEMATICS GRADE 10

Sub-Strand 1.1: Real Numbers — Teacher Reference (Pre-filled)

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SUMMARY TABLE: CORE MATHEMATICS GRADE 10

Sub-Strand	Sub-Strand 1.1: Real Numbers
Driving Question	DRIVING QUESTION: Why can we never write some everyday numbers down completely — and how do mathematicians and Kenyan professionals work accurately with numbers that go on forever?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 What Kind of Number Is That? (Predict & Classify)	Students examined a set of numbers drawn from Kenyan everyday life — market prices (KSh 60.50), maize bag weights (160 kg), temperature readings (34.5 °C), circular land measurements involving π , and diagonal fence lengths involving $\sqrt{2}$. They observed that some numbers terminated neatly on a calculator or paper, while others continued without end and without a repeating block, no matter how many decimal places were computed.	Students learned that numbers fall into two broad camps based on their decimal behaviour: those that can be written down completely (terminating decimals and whole numbers) and those that cannot (non-terminating, non-repeating decimals such as π and $\sqrt{2}$). This distinction is the foundational crack in the number system that the entire sub-strand will explore. Pedagogical note: Most students arrive believing all numbers can eventually be 'finished' — this lesson is designed to	The inability to write certain numbers completely is not a failure of mathematics — it is a structural feature of the real number system. Numbers like π arise naturally in Kenyan contexts (calculating the circumference of a circular borehole in Turkana, the area of a round ugali sufuria) yet can never be expressed as a ratio of two integers. This sets up the driving question: if professionals — engineers, pharmacists, surveyors in the Rift Valley — must work with such numbers daily, how do

		create productive cognitive dissonance. Preserve that dissonance; do not resolve it prematurely.	they do so accurately? Students should leave understanding that 'going on forever' is a property to be classified and managed, not ignored.
Lesson 2 Sorting the Number World: Classifying Real Numbers from Kenyan Contexts	Students sorted a rich card set of numbers sourced from Kenyan contexts — population counts (natural numbers), temperature drops below zero in Nanyuki at night (negative integers), sugar prices per gram (fractions/rationals), fabric lengths involving $\sqrt{3}$ (irrationals), and altitude readings for Mt. Kenya (integers). They physically arranged cards into nested hoops labelled Natural Numbers, Integers, Rational Numbers, and Real Numbers, observing which sets contained others and where irrational numbers sat outside the rational hoop but still inside Real Numbers.	Students learned the formal nested-set structure of the real number system: $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$, with irrational numbers ($\mathbb{Q}^c$) filling the 'gaps' between rationals on the number line. They learned precise set notation and understood that every real number is either rational or irrational — never both. Pedagogical note: Emphasise the subset arrows with concrete examples: every counting number (1 maize cob, 2 goats) is also an integer, also a rational, also a real — but not every real is a natural number. Use the nested hoops as a persistent classroom anchor throughout the sub-strand.	The nested structure explains why a Kenyan pharmacist dispensing 0.75 g of a drug and a land surveyor measuring $\sqrt{5}$ metres of fencing are both working within the same real number system, just in different sub-regions of it. The completeness of $\mathbb{R}$ — the fact that it contains all rationals AND all irrationals — is what makes it the correct number system for measuring any continuous quantity in the physical world, from the depth of Lake Victoria to the wavelength of light at a Nairobi hospital.
Lesson 3 Rational or Irrational? Sorting the Number Family	Students investigated the decimal expansions of a curated list of numbers using long division and calculators: simple fractions ($3/4$, $1/3$, $5/6$), square roots of perfect squares ($\sqrt{9}$, $\sqrt{16}$), square roots of non-perfect squares ($\sqrt{2}$, $\sqrt{5}$, $\sqrt{7}$), and named constants (π). They recorded whether each decimal terminated, recurred, or neither, and attempted to express each as a fraction p/q where p and q are integers and $q \neq 0$.	Students learned the formal definitions: a rational number is any number expressible as p/q ($p, q \in \mathbb{Z}$, $q \neq 0$), terminating decimals are rational (they can be written as fractions with powers-of-10 denominators), recurring decimals are rational (they can be converted to fractions via the algebraic 'multiply-and-subtract' method), and irrational numbers are those whose decimal expansions are non-terminating AND non-repeating — they cannot be expressed as p/q for any integers p, q . Pedagogical note: The recurring-decimal-to-fraction conversion (e.g. let $x = 0.\overline{3}$, then $10x = 3.\overline{3}$, subtract to get $9x = 3$, so $x = 1/3$) is a high-yield procedure — allocate sufficient practice time and cold-call at least four students to narrate each step aloud.	The p/q test is the professional mathematician's tool for classifying any number. When a Kenyan engineer approximates π as $22/7$, she is deliberately replacing an irrational number with a nearby rational one to make computation tractable — but she knows $22/7 \neq \pi$. Understanding the boundary between rational and irrational is what allows professionals to choose approximations consciously and quantify the resulting error, directly answering the driving question.
Lesson 4 Placing Numbers on the Line: Can We Always Find the Exact Spot?	Students constructed number lines and used geometric methods (right-triangle constructions on grid paper) to locate irrational numbers including $\sqrt{2}$, $\sqrt{3}$, and $\sqrt{5}$ at their exact positions. They observed that even though these numbers cannot be written as terminating or recurring decimals,	Students learned that the real number line is continuous and complete: every real number — rational or irrational — corresponds to exactly one point, and every point corresponds to exactly one real number (the completeness property of $\mathbb{R}$). Irrational numbers are not 'missing' or	A Kenyan land surveyor in Kajiado marking out a square plot of exactly 1 hectare ($10\,000\text{ m}^2$) needs the exact side length: $\sqrt{10\,000} = 100\text{ m}$ (rational, fine). But a plot of 2 hectares has side $\sqrt{20\,000} = 100\sqrt{2}\text{ m}$ — an irrational length that is nonetheless a precise, real, physical distance on the

	a precise geometric point for each exists on the line — visible, unambiguous, and distinct from all nearby rational approximations.	'approximate' — they are exact locations. They also learned to estimate the positions of irrationals by identifying the two consecutive integers they fall between (e.g. $\sqrt{10}$ lies between 3 and 4 because $3^2 = 9 < 10 < 16 = 4^2$). Pedagogical note: The compass-and-ruler location of $\sqrt{2}$ is a powerful concrete proof that irrational numbers exist as exact points — refer back to this construction whenever students lapse into treating irrationals as 'just approximations.'	ground. The number line model confirms that such distances exist and can be located exactly, even if they cannot be written as terminating decimals.
Lesson 5 Approximating Irrational Numbers: Working Accurately with Infinite Decimals	Students used calculators and printed decimal expansion strips of $\sqrt{2}$, $\sqrt{3}$, and π to explore what happens when these numbers are rounded to 1, 2, 3, and 4 decimal places. They computed the absolute error at each level of rounding and examined real Kenyan professional contexts — a Mombasa architect's blueprint, a Kericho tea-factory mixing ratio, a pharmacist's dosage — to judge which degree of approximation was 'good enough' for each purpose.	Students learned the procedures for rounding and truncating irrational (and rational) numbers to a specified number of decimal places or significant figures, and how to compute and interpret absolute error (true value – approximation). They learned that the choice of approximation precision is context-dependent: a primary-school textbook may write $\pi \approx 3.14$, an engineer uses $\pi \approx 3.14159$, and a physicist may use far more digits — but all are deliberate, justified choices. Pedagogical note: Require students to state both the approximation AND the maximum error in every calculation this lesson — build the habit of error-awareness that underpins the driving question answer.	The driving question asks how professionals work accurately with numbers that go on forever. This lesson supplies the core answer: by choosing a rational approximation that is precise enough for the task, and by knowing the size of the resulting error. A Kenyan civil engineer designing the circular base of a water tank in Eldoret uses $\pi \approx 3.1416$ and knows her circumference calculation is accurate to within a fraction of a millimetre — more than sufficient for concrete formwork. Accuracy is not about writing the whole number; it is about controlling the error.
Lesson 6 Operating on Real Numbers: BODMAS in the Market and the Sufuria	Students evaluated multi-step numerical expressions drawn from Kenyan market and kitchen contexts — e.g. total cost of sukuma wiki, tomatoes, and cooking oil with a discount applied; volume of a cylindrical sufuria using π; floor area combining rectangular and semicircular sections. Expressions mixed integers, fractions, decimals, and irrational numbers (π, $\sqrt{2}$). Students recorded their step-by-step working and compared answers when BODMAS was applied correctly versus when it was violated.	Students learned that the order of operations (Brackets, Orders/powers, Division, Multiplication, Addition, Subtraction — BODMAS) applies universally across ALL real numbers — rational and irrational alike — and that violating the order produces wrong answers that could have real-world consequences (incorrect drug dosages, under-ordered building materials). They practised simplifying expressions containing nested brackets, integer exponents, and irrational numbers, leaving surds (e.g. $3\sqrt{2}$) in exact form where appropriate. Pedagogical note: A common error is left-to-right processing of addition before multiplication. Use the market-bill context to motivate correctness	BODMAS is not an arbitrary rule — it is a convention that makes mathematical expressions unambiguous for everyone who reads them, from a Form 4 student in Kisumu to a quantity surveyor in Thika. Because real numbers include both the tidy rationals of daily commerce and the infinite irrationals of geometry and science, a single consistent set of operational rules is essential. This lesson demonstrates that the real number system is closed and well-behaved under the standard operations, reinforcing students' growing model of $\mathbb{R}$ as a coherent, complete system.

		— overpaying at Gikomba Market because of a BODMAS error is a tangible consequence students connect with.	
Lesson 7 Surds: Simplifying and Operating on Irrational Square Roots	Students manipulated physical 'square tiles' (cut-out unit squares) to build squares of area 2, 3, 5, and 8, observing that only perfect-square areas produced whole-number side lengths. They then used the multiplication and simplification rules for surds — $\sqrt{ab} = \sqrt{a} \cdot \sqrt{b}$ and $\sqrt{a/b} = \sqrt{a} / \sqrt{b}$ — to simplify expressions such as $\sqrt{50}$, $\sqrt{72}$, and $2\sqrt{3} \times 5\sqrt{3}$ by hand, checking results on calculators. They also added and subtracted like surds ($3\sqrt{2} + 5\sqrt{2} = 8\sqrt{2}$) and recognised that unlike surds ($\sqrt{2} + \sqrt{3}$) cannot be combined.	Students learned the definition of a surd (an irrational root that cannot be simplified to a rational number), the rules for multiplying, dividing, and simplifying surds, the process of rationalising a denominator (multiplying numerator and denominator by the conjugate surd to eliminate irrationals from the denominator), and the condition for adding/subtracting surds (like surds only). Pedagogical note: Rationalising the denominator is procedurally straightforward but conceptually important — emphasise that $1/\sqrt{2}$ and $\sqrt{2}/2$ are equal; the second form is preferred because all irrational complexity sits in the numerator, making further computation cleaner. Have students verify numerically on calculators.	Surds appear whenever Kenyan engineers, architects, or scientists compute exact lengths, areas, or forces involving square roots — the diagonal of a square plot, the hypotenuse of a right-triangular roof truss, the resonant frequency of a bridge. Working with surds in exact form (rather than decimal approximations) eliminates accumulated rounding error across multi-step calculations. This lesson equips students with the algebraic toolkit for exact irrational arithmetic, a key component of the driving-question answer.
Lesson 8 Absolute Value: Distance from Zero — and the Final Explanation of the Real Number System	Students plotted positive and negative real numbers on a number line (temperatures above and below 0 °C in Nanyuki, profit and loss figures for a Nairobi small business, altitude above and below sea level at the coast versus the highlands) and measured, in each case, the physical distance of each number from zero — always obtaining a non-negative result. They evaluated expressions such as -7, 3.5, $\sqrt{2} - 2$, and x for given values of x, and solved simple absolute-value equations ($x = 5 \rightarrow x = 5$ or $x = -5$).	Students learned the formal piecewise definition of absolute value: $x = x$ if $x \geq 0$; $x = -x$ if $x < 0$. They learned that x represents distance from zero and is always non-negative; that $x - a$ represents the distance between x and a on the number line; and that absolute-value equations have two solutions (one positive, one negative branch) unless the right-hand side is zero. They consolidated understanding of the entire real number system — natural numbers, integers, rationals, irrationals, surds, approximation, and order of operations — in a final synthesis activity. Pedagogical note: The synthesis activity is the culmination of the sub-strand. Ask students to return to the driving question explicitly and construct a written answer in their own words before the class co-constructs a model answer together. This metacognitive closure is essential for long-term retention.	Absolute value completes the real number system toolkit by providing a way to measure magnitude independently of direction or sign — critical for error analysis ($\text{approximate value} - \text{exact value}$ is an absolute error), for distance problems in navigation and surveying, and for any context where only 'how far?' matters, not 'which direction?'. The driving question is now fully answerable: some everyday numbers (irrationals like π and $\sqrt{2}$) cannot be written down completely because they are non-terminating and non-repeating — a structural property of $\mathbb{R}$. Kenyan professionals handle them accurately by (a) using justified rational approximations with known error bounds, (b) applying surd algebra for exact intermediate results, and (c) using consistent operational rules (BODMAS, absolute value) across the entire real number system.